

Exploring Catalan Numbers and Their Strikingly Distinct Applications

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1 Introduction

The Catalan numbers are a set of nonnegative integers used in various combinatorics problems. Though named after the French-Belgian mathematician Eugène Charles Catalan, the sequence was earlier discovered by Leonhard Euler while exploring the triangulation of polygons. Indeed, the Catalan numbers can be used to count the number of triangulations of a polygon.

1.0.1 Example

How many triangulations are there of a polygon? To understand this problem, we define a triangulation.

1.0.2 Definition

A triangulation is a dissection of a polygon into triangles with all vertices being those of the original polygon, such that none of the sides intersect.

First, verify that every polygon actually has a triangulation.

1.0.3 Claim

Every polygon has a triangulation.

1.0.4 Proof

We proceed by strong induction on n , the number of vertices in the polygon.

Base Case: $n = 3$. This case is trivial as the polygon is a triangle itself.

Inductive Step: For $n > 3$, assume that the theorem holds for all $k < n$. Now, let P be a polygon with n vertices. P must have a diagonal, so we can draw one which splits P into two adjacent polygons, M and N . M must have at least one vertex that is not shared with N . Similarly,

N must have at least one vertex that is not shared with M . Thus, both M and N have fewer than n vertices. By our inductive hypothesis, both have triangulations, meaning that P also has a triangulation.

Now, let's delve into the combinatorial problem of how many triangulations there actually are of an n -gon. Let us begin with a recursive relation. Consider an arbitrary polygon Z with $n + 2$ sides. By Claim 1.0.3, Z has a triangulation. Denote the number of ways to triangulate it as C_n . Now, fix a side of Z called AB . In any triangulation, AB forms a triangle with another vertex of Z ; let it be C . $\triangle ABC$ is in the triangulation. This triangle cuts the original polygon into three polygons: $\triangle ABC$, a polygon F with side AC , and a polygon G with side BC .

Without loss of generality, let the polygon F have $k + 2$ sides and k triangles in its triangulation. Then the polygon G will have $n - k + 1$ sides and $n - k - 1$ triangles in its triangulation, where k can change from 0 to $n - 1$. Thus, there are

$$\sum_{k=0}^{n-1} C_k C_{n-1-k}$$

ways to triangulate the originally defined polygon. This is the recurrence that defines the Catalan numbers.

Now, consider another problem.

1.0.5 Example

Given an even number of people seated at a round table, how many ways can simultaneous handshakes occur without anyone's arms crossing?

1.0.6 Claim

This satisfies the Catalan recurrence.

1.0.7 Proof

Let the people be numbered

$$1, 2, 3, \dots, 2n$$

around the table. If person 1 shakes hands with an odd numbered person, then there will be an odd number of people in-between the pair. Given that no crossing handshakes are allowed, among the odd number of people, there will always be a lone person without a pair. Thus, person 1 must shake hands with an even numbered person.

Now, we can split the problem into cases and quantify them using C_n . If person 1 shakes hands with person 2, then the remaining people have C_{n-1} ways to shake hands. If person 1 shakes hands with person 4, then person 2 can shake hands with person 3 in C_1 ways, and the other people can shake hands together in C_{n-2} ways. What do you notice now? This clearly satisfies the same recurrence $\sum_{k=0}^{n-1} C_k C_{n-1-k}$.

1.0.8 Theorem

The Catalan recurrence can be succinctly defined by the non-recursive relation

$$C(x) = \sum_{n=0}^{\infty} \binom{2n}{n} \frac{x^n}{n+1}.$$

1.0.9 Proof

The generating function for the Catalan numbers is defined by the power series

$$C(x) = \sum_{n=0}^{\infty} C_n x^n.$$

However, we obtain a different generating function if we expand the Catalan recurrence relation:

$$C(x) = 1 + x \cdot C(x)^2.$$

Rewriting as $xC(x)^2 - C(x) + 1 = 0$ and solving for $C(x)$ using the quadratic formula, we get

$$\frac{1 \pm \sqrt{1 - 4x}}{2x}.$$

Regarding the $\pm$, we must take the $-$ in order to satisfy $C_0 = \lim_{x \rightarrow 0} C(x) = 1$. Since the power series is centered at 0, its coefficients are equal to the Catalan numbers, and thus the square root term $\sqrt{1 - 4x}$ can be rewritten as the power series

$$\sum_{n=1}^{\infty} \frac{1}{2(2n-1)} \binom{2n}{n} x^{n-1}.$$

A simple substitution yields

$$C(x) = \sum_{n=0}^{\infty} \binom{2n}{n} \frac{x^n}{n+1},$$

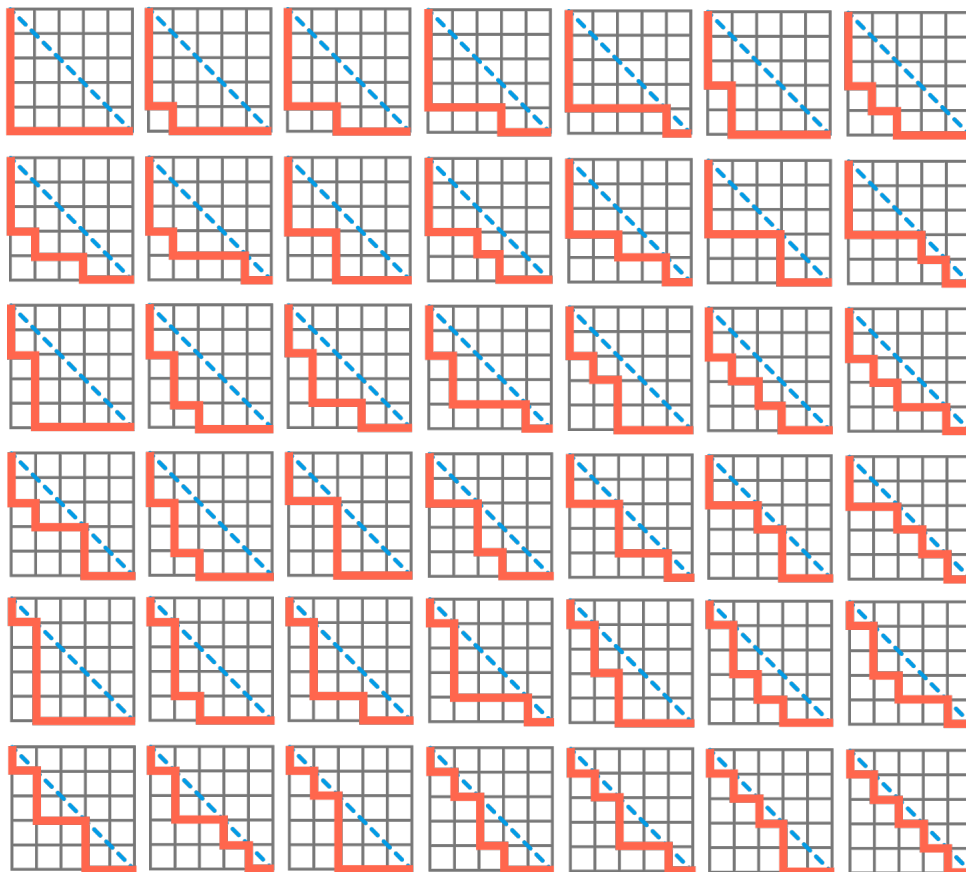
so we are done.

1.0.10 Exercise

Verify the statement above using induction.

1.0.11 Example

A well-known problem involving the non-recursive definition of the Catalan sequence is block walking. We aim to find a closed formula for the number of paths of length $2n$ from one corner of an $n \times n$ square to the opposite one without crossing the diagonal connecting them (solutions for $n = 5$ shown below).



First, we can calculate the total number of paths (ignoring the restriction for now). There must be n vertical steps and n horizontal steps, so out of the $2n$ steps, choosing any n of them to be vertical determines a unique path. Therefore, there are $\binom{2n}{n}$ total paths.

Then, define a “bad” path as one that crosses the main diagonal and touches the next diagonal up/down. If we reflect the invalid portion of the path, there are now $n + 1$ vertical steps and $n - 1$ horizontal steps. Thus the number of bad paths can be expressed as

$$\binom{n - 1 + n + 1}{n - 1} = \binom{2n}{n - 1} = \binom{2n}{n + 1}.$$

Moreover, the number of desired paths is

$$\binom{2n}{n} - \binom{2n}{n + 1}.$$

Using induction to verify that expression this is equivalent to all the other Catalan formulas is left as an exercise to the reader.

1.0.12 Exercise

Consider a bijection of this to handshaking; correspond each person shaking hands with an even number but not an odd number to walking along the black line but not the red line.

1.0.13 Remark

The idea of taking a bijection considered in the exercise above can be applied to all Catalan-type recurrence relations.

1.1 Problems

1. Compute C_6 and C_7 using both the recursion and the explicit formula.
2. How many ways are there to arrange n open brackets “[” and n closed brackets “]” such that when read left to right, the number of closed brackets is less than or equal to the number of open brackets?

2 The Hausdorff Moment Problem

We begin with a definition.

2.0.1 Definition

d -dimensional Catalan numbers $C_d(n)$ are defined by

$$\prod_{p=0}^{d-1} \frac{p!}{(n+p)!} (dn)! \quad [1]$$

where $2 \leq d \in \mathbb{Z}$, and $0 \leq n \in \mathbb{Z}$.

2.1 Lemma

The integral representation of the Catalan sequence is

$$C_n = \frac{1}{2\pi} \int_0^4 x^n \sqrt{\frac{4-x}{x}} dx.$$

The following section will provide a walk-through of the extensive proof of the key theorem below:

2.1.1 Theorem

These $C_d(n)$'s are the n th Hausdorff power moments of positive functions $W_d(x)$ for $x \in [0, d^d]$.

Catalan numbers that can be associated with a dimension d allow for d -dimensional extensions of the theorems for the two-dimensional Catalan numbers explored in the Introduction. Note that by [1], we have that $C_d(0) = 1$ for all d . We seek the solutions of the Hausdorff Moment Problem

$$\int_0^{R(d)} x^n W_d(x) dx = C_d(n),$$

where $R(d)$ will be the upper edge of $W_d(x)$'s support:

$$R(d) = \lim_{n \rightarrow \infty} (C_d(n))^{\frac{1}{n}} = d^d.$$

2.1.2 Claim

The $W_d(x)$ that we seek in the Hausdorff Moment Problem is positive.

2.1.3 Proof

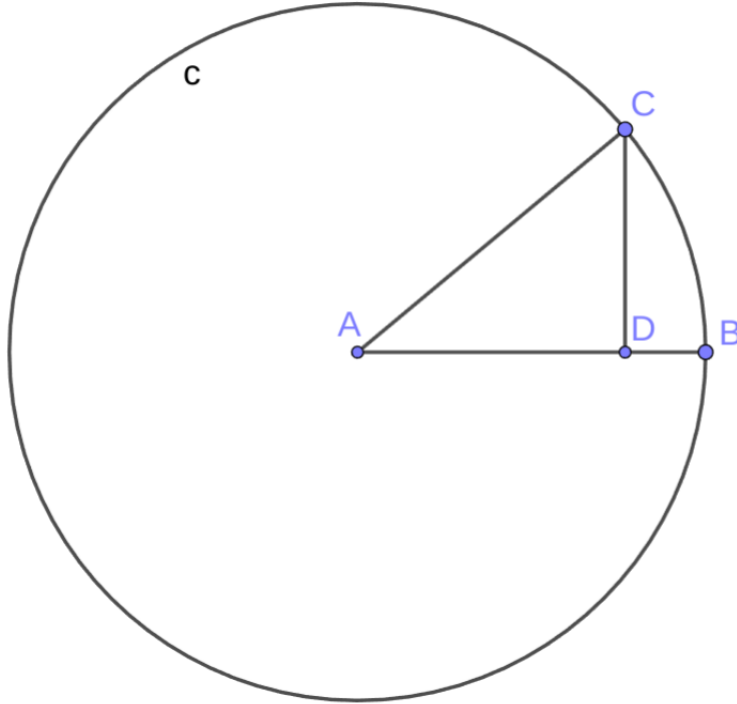
By the Gauss-Legendere multiplication formula for the gamma function on [1] and for complex z such that $z = n + 1$, we have that

$$C_d(z - 1) = (2\pi)^{1-d/2} \cdot d^{\frac{1-2d}{2}} \cdot \prod_{k=0}^{d-1} k! \cdot (d^d)^z \cdot \prod_{j=0}^{d-1} \frac{\Gamma(n + j + 1)/d}{\Gamma(z + j)}$$

is the Mellin transform of $W_d(x)$. From here, we note that j has a Mellin transform that indicates its proportionality to the standard beta distribution over $[0, 1]$. Looking at the $C_d(z - 1)$ as the product of d individual terms, we can see that $W_d(x)$ is continuous and thus positive, because it is a d -fold Mellin transformation of positive, continuous functions over $[0, 1]$. Then, we note that the Hausdorff Moment Problem is dependent on $\frac{x}{d^d}$, a d -fold Mellin convolution. Executing this Mellin convolution is done by identifying $W_d(x)$ by the Meijer G -function, and then converting it to hypergeometric form. This gives us $R(d) = d^d$.

From here, computing the coefficients $c_j(d)$ using the integral representation of the Catalan numbers along with a density argument gives that the $W_d(x)$ give solutions, as desired.

3 A Surprising Geometric Result



Consider circle c with radius $r = AB = AC$. Let the length of CD be 1. Then,

$$AD = \sqrt{r^2 - 1}$$

and

$$DB = r - \sqrt{r^2 - 1}.$$

Seeking to find the length of DB , we can try plugging in some small values of r . For example, $r = 5$ gives $DB = 0.101020514433643803\dots$, and if you look closely, you'll spot the numbers 1, 1, 2, 5, 14, and a number approaching 42 hiding in the decimal expansion. These are the first six Catalan numbers! But wait, there's more... Plugging in $r = 50,000,000,000,000,000$, we get:

[illegible]

Between the zeros lie Catalan numbers galore! Why does this work?

Think about generating functions. When we use the Taylor series expansion of the Catalan recurrence relation, $\frac{1 - \sqrt{1 - 4x}}{2x}$, we can let $r = 5 \cdot 10^{n-1}$. Plugging into our expression $r - \sqrt{r^2 - 1}$ and expanding with the Taylor series will imply the problem; the rest is left as an exercise to the reader.

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